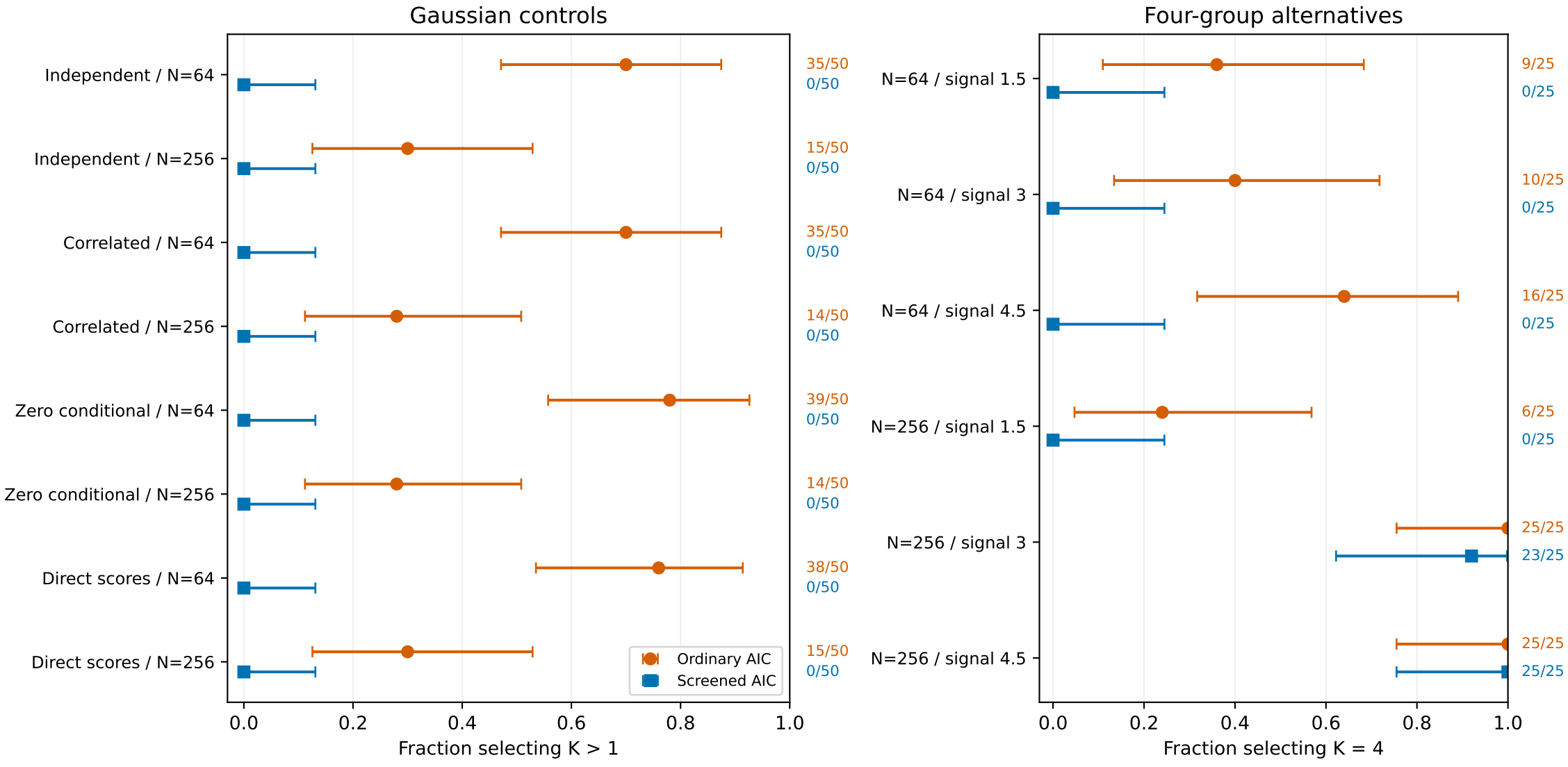
Predictive screening reduces partitioning, with a recovery tradeoff
95% simultaneous exact intervals across 28 prespecified rates
Gaussian controls
Four-group alternatives
Independent / N=64
Independent / N=256
Correlated / N=64
Correlated / N=256
Zero conditional / N=64
Zero conditional / N=256
Direct scores / N=64
Direct scores / N=256
35/50
0/50
15/50
0/50
35/50
0/50
14/50
0/50
39/50
0/50
14/50
0/50
38/50
0/50
15/50
0/50
Ordinary AIC
Screened AIC
0.0
0.2
0.4
0.6
0.8
1.0
Fraction selecting K > 1
N=64 / signal 1.5
N=64 / signal 3
N=64 / signal 4.5
N=256 / signal 1.5
N=256 / signal 3
N=256 / signal 4.5
9/25
0/25
10/25
0/25
16/25
0/25
6/25
0/25
25/25
23/25
25/25
25/25
0.0
0.2
0.4
0.6
0.8
1.0
Fraction selecting K = 4